CHAPTERS	HEADINGS	PAGE.NO
I	INTRODUCTION	
II	BASIC CONNCEPTS	
III	INFOREMATION THEORY	
IV	FORECASTING	
V	EXAMPLES	

CHAPTER -I

INTRODUCTION

Information Theory and Forecasting are important areas of mathematics and statistics that deal with information and data. Information Theory is the mathematical study of information, its measurement, transmission, storage and processing. It helps us to understand the amount of information contained in an event and the uncertainty associated with it. Important concepts in Information Theory include measure of information, entropy, communication systems, channel probabilities, joint and conditional entropies, mutual information and encoding.

Forecasting is the process of estimating or predicting future events or values by using past and present observations. It is widely used in business, economics, statistics, industry and other fields for making decisions and planning for the future. Forecasting mainly deals with time-series data and its components such as trend, seasonal, cyclical and irregular variations. Important forecasting methods include smoothing methods, trend projection, causal methods and qualitative methods.

Thus, Information Theory helps in understanding and handling information, whereas Forecasting helps in analysing data and predicting future values. Both are useful tools for mathematical analysis, statistical decision-making and data-based prediction

This project entitled **“A STUDY ON INFORMATION THEORY AND FORECASTING”** consists Five chapters.

In the first chapter, **Introduction**, we introduce the basic concepts of Information Theory and Forecasting.

In the second chapter, Introduction, we recall some **Basic Concepts** of Information Theory and Forecasting which are used in the subsequent chapters.

In the third chapter, we study the concept of **Information Theory.**

In the forth chapter, we study the concept of **Forecasting and Time Series.**

In the fifth chapter, we study the concept of **Examples** in information theory and forecasting .

CHAPTER-II

BASIC CONCEPTS

INFORMATION THEORY

Definition: **Information Theory** is the mathematical study of measuring, storing and communicating information.

Example: Sending a message through a communication channel.

Formula: No specific formula.

A Measure of Information

Definition: The amount of information obtained from an event is called the measure of information.

Formula:

I(x)=-log_2(p(x))

Example: If p(x)= $\frac{1}{2}$,

I(x)=$-log_2 \frac{1}{2}$=1 bit

Entropy – The Expected Information

Definition: Entropy is the average amount of information contained in a random variable.

Formula:

$$H(x) = -\sum_{i=1}^{n} p_i \, log_2 p_i$$

Example: For a fair coin $H(x) = 1 \; bit$

Entropy as a Measure of Uncertainty

Definition: Entropy measures the uncertainty or randomness of a random variable.

Formula:

$$H(x) = -\sum p(x) log_2 p(x)$$

Example: A fair coin has more uncertainty than a coin that always gives Head.

Some Properties of Entropy Function

Definition: Entropy has certain mathematical properties that describe the uncertainty of a random variable.

Important formula/property: $H(x) \geq 0$

For maximum uncertainty: $H(x) \leq log_2 n$

where □ is the number of possible outcomes.

The Communication System

Definition: A communication system transmits information from a source to a destination through a channel.

Example:

$$Source \rightarrow Transmitter \rightarrow Channel \rightarrow Recri$$

Channel Probabilities

Definition: Channel probability is the probability of receiving a particular output when a particular input is transmitted.

Formula:

$$P(y = y_i | x = x_i)$$

Example: $P(y = 1 | x = 0)$

means probability of receiving 1 when 0 is transmitted.

Mutual Information

Definition: Mutual information measures how much information one random variable gives about another.

Formula:

$$I(x;y) = H(x) - H(x|y)$$

Also,

$$I(x;y) = H(y) - H(y|x)$$

Example: If knowing □ reduces the uncertainty of □, the reduction is mutual information.

Encoding

Definition: Encoding is the process of converting information into a suitable code for transmission or storage.

Example: Converting A into binary 01000001.

Formula: In basic encoding, the exact formula depends on the coding method used.

FORECASTING

Definition: Forecasting is the process of predicting future events or values using past and present data.

Example: Predicting next year's sales using previous years' sales data.

Forecasting Methods

Definition: Forecasting methods are techniques used to estimate future values.

Examples: Moving Average, Exponential Smoothing and Trend Projection.

Formula: Depends on the forecasting method used.

Components of a Time Series

Definition: The different types of variations present in time-series data are called components of a time series.

Components:

1. Trend (T)
2. Seasonal Variation (S)
3. Cyclical Variation (C)
4. Irregular Variation (I)

Formula:

$$Y = T \times S \times C \times I$$

Smoothing Methods in Forecasting

Definition: Smoothing methods reduce random fluctuations and show the underlying pattern in time-series data.

Example: Moving Average.

Formula:

Moving Average=$\frac{Sum\ of\ observations}{Number\ oF\ observations}$

Trend Projection in Forecasting

Definition: Trend Projection predicts future values by extending the existing trend.

Formula:$Y = a + bx$

Where,

Y = Trend value

a = Constant

b = Slope

x = Time

Example: Predicting future sales from past sales trend.

Seasonal Components in Forecasting

Definition: Seasonal variation is a regular variation that repeats during a particular period.

Example: Ice cream sales increase during summer.

Formula:

Seasonal Index=$\frac{seasonal\ average}{Grand\ Average} \times 100$

Causal Forecasting Method

Definition: Causal forecasting predicts one variable based on its relationship with another variable.

Formula:

$$Y = a + bX$$

Example: Predicting sales Y based on advertising expenditure X.

Qualitative Methods of Forecasting

Definition: Qualitative forecasting uses expert opinion, experience and judgement to predict future events.

Example: Experts predicting the demand for a new product.

CHAPTER-III

INFORMATION THEORY

Claude Shannon in his landmark 1948 paper "A Mathematical Theory of Communication," that deals with the quantification, storage, and transmission of information. It provides the theoretical foundation for modern digital communication systems, data compression, and error correction.

At its core, information theory answers two fundamental questions:

1. How much can a message be compressed? governed by the concept of entropy, which measures the average uncertainty or information content of a source.
2. How fast can data be transmitted reliably over a noisy channel? governed by the concept of channel capacity.

Before Shannon's work, communication engineering was largely empirical. Shannon introduced a rigorous mathematical model that separates the source (what is being communicated), the channel (the noisy medium through which it travels), and the receiver (which must decode the message). This separation allows engineers to design source coding (compression) and channel coding (error correction) independently a principle known as the source-channel separation theorem.

Key concepts explored in this project include:

- Measure of Information – quantifying the information content of an event based on its probability of occurrence; rarer events carry more information.
- Entropy – the expected (average) information content of a source, representing the minimum number of bits needed to represent a symbol without loss.

- Joint and Conditional Entropy – extending entropy to multiple correlated random variables, essential for analyzing communication over noisy channels.
- Mutual Information – the amount of information one random variable contains about another, used to measure how much information survives transmission through a noisy channel.
- Channel Capacity – the maximum rate at which information can be reliably transmitted over a channel, as established by Shannon's Noisy Channel Coding Theorem.
- Encoding – practical techniques (e.g., Huffman coding) that approach the theoretical entropy limit, enabling efficient and lossless data compression.

A Measure of Information

Before information can be measured mathematically, we need a way to quantify how much "information" is conveyed when a particular event occurs. The central idea introduced by Shannon is simple but powerful: the amount of information gained from an event is inversely related to its probability of occurrence.

In other words:

- An event that is certain to happen (probability = 1) gives no new information when it occurs, since we already expected it.
- An event that is rare or unexpected (low probability) gives a large amount of information when it occurs, because it resolves a great deal of uncertainty.

Mathematical Definition

If an event x occurs with probability P(x), the amount of information associated with the occurrence of that event is defined as:

$I(x) = \log_2 (1 / P(x)) = -\log_2 P(x)$

where:

I(x) is the information content (also called self-information) of the event, measured in bits (when log base 2 is used).

P(x) is the probability of occurrence of the event.

Properties of the Information Measure

1. $I(x) \geq 0$ information is always non-negative (since $0 \leq P(x) \leq 1$).
2. $I(x) = 0$ when $P(x) = 1$ a certain event carries no information.
3. I(x) increases as P(x) decreases rarer events carry more information.
4. Additivity for two independent events x and y, $I(x , y) = I(x) + I(y)$, since $P(x , y) = P(x) \cdot P(y)$ and logarithms convert products into sums.

Example 1: Fair Coin Toss

A fair coin has P(Heads) = P(Tails) = 0.5.

$I(\text{Heads}) = -\log_2 (0.5) = 1$ bit.

→ Each outcome of a fair coin toss carries exactly 1 bit of information.

Example 2: Biased Coin

Suppose a biased coin has P(Heads) = 0.9, P(Tails) = 0.1.

$I(\text{Heads}) = -\log_2 (0.9) \approx 0.152$ bits (low information heads was expected)

$I(\text{Tails}) = -\log_2 (0.1) \approx 3.32$ bits (high information tails was a surprise)

→ This shows unexpected outcomes carry more information than expected ones.

Example 3: Rolling a Fair Die

Each face has P = 1/6.

$I(\text{any face}) = -\log_2 (1/6) \approx 2.585$ bits.

Example 4: Weather Report

If a desert region reports "It rained today" (P = 0.01), the information content is:

I = −log□ (0.01) ≈ 6.64 bits (highly informative rare event).

If it reports "It was sunny today" (P = 0.9), the information content is:

I = −log□ (0.9) ≈ 0.152 bits (barely informative expected event).

Example 5: Independent Events

Tossing two fair coins together:

P (Heads, Heads) = 0.5 × 0.5 = 0.25

I (Heads, Heads) = −log□ (0.25) = 2 bits = I(Heads) + I(Heads) = 1 + 1 □ (confirms additivity)

Entropy — The Expected Information

While self-information I(x) tells us how much information a single event carries, most real sources produce a sequence of symbols, each with its own probability. Entropy extends this idea by measuring the average (expected) information content produced by a source over all its possible symbols. It represents the fundamental lower bound on how much a source can be compressed without loss of information.

Mathematical Definition

For a discrete random variable (source) X with possible outcomes x□, x□, ..., x⍰ and corresponding probabilities P(x□), P(x□), ..., P(x⍰), the entropy H(X) is defined as:

$H(X) = \Sigma P(x_i) \cdot I(x_i) = -\Sigma P(x_i) \log_{□} P(x_i)$ (summed over i = 1 to n)

Entropy is measured in bits per symbol and represents:

- The average uncertainty associated with predicting the source's output, and
- The minimum average number of bits required to encode each symbol from that source.

Key Points

- Entropy is a weighted average of the self-information of all possible outcomes, weighted by their probabilities.
- H(X) = 0 only when one outcome has probability 1 (no uncertainty the source is completely predictable).
- H(X) is maximum when all outcomes are equally likely maximum uncertainty means maximum average information.
- For a source with n equally likely symbols, maximum entropy = log□(n) bits/symbol.

Example 1: Fair Coin (Binary Source)

P(Heads) = P(Tails) = 0.5

H(X) = −[0.5 log□0.5 + 0.5 log□0.5] = −[0.5(−1) + 0.5(−1)] = 1 bit/symbol

→ Maximum entropy for a binary source — each toss is maximally unpredictable.

Example 2: Biased Coin

P(Heads) = 0.9, P(Tails) = 0.1

H(X) = − [0.9 log□0.9 + 0.1 log□0.1]

= − [0.9(−0.152) + 0.1(−3.32)]

= 0.137 + 0.332 = 0.469 bits/symbol

→ Much lower than 1 bit, because the outcome is largely predictable (mostly Heads), so less new information is generated on average.

Example 3: Fair Die

Six equally likely outcomes, P = 1/6 each.

H(X) = log□ (6) ≈ 2.585 bits/symbol

→ This matches the maximum entropy formula log□(n) for n = 6 equally likely outcomes.

Entropy as a Measure of Uncertainty

Entropy is not only the expected information of a source — it is equally, and perhaps more intuitively, understood as a measure of uncertainty or randomness associated with a random variable. The more unpredictable a source's behavior , the higher its entropy; the more predictable it is, the lower its entropy. This dual interpretation (expected information ≡ uncertainty) is one of the most powerful insights of information theory.

Conceptual Basis

Before an event occurs, there is uncertainty about which outcome will happen. Once the event occurs, that uncertainty is resolved, and in the process, information is gained. Therefore:

Uncertainty before the event = Information gained after the event

This is why entropy H(X) can be interpreted in two equivalent ways:

1. The average information produced by the source per symbol.
2. The average uncertainty an observer has about the source's next symbol, before it is revealed.
3. Why Entropy Reflects Uncertainty
   - If a source is completely predictable (one outcome has probability 1), there is no uncertainty about the outcome → H(X) = 0.
   - If a source is completely random (all outcomes equally likely), there is maximum uncertainty about the outcome → H(X) is maximum.
   - Any distribution between these extremes has intermediate entropy, reflecting partial predictability.

This relationship is best visualized using a binary source with P(1) = p and P(0) = 1−p, plotted as H(p) vs. p — a curve that starts at 0 (p=0), rises to a peak of 1 bit at p=0.5, and falls back to 0 (p=1). This is called the binary entropy function:

$H(p) = -p \log_2 p - (1-p) \log_2(1-p)$

Example 1: Certain Outcome (No Uncertainty)

A loaded die always lands on 6: P(6) = 1, P(others) = 0.

H(X) = −1·log□ 1 = 0 bits.

→ Zero uncertainty — you already know the outcome, so entropy is zero.

Example 2: Maximum Uncertainty (Fair Coin)

P(Heads) = P(Tails) = 0.5

H(X) = 1 bit (maximum possible for a binary source).

→ You have absolutely no way to predict the next toss — maximum uncertainty.

Example 3: Partially Predictable Source

A weighted coin: P(Heads) = 0.8, P(Tails) = 0.2

H(X) = − [0.8log□0.8 + 0.2log□0.2] = − [0.8(−0.322) + 0.2(−2.322)]

= 0.257 + 0.464 = 0.722 bits

→ Less than 1 bit some uncertainty remains, but it's reduced because Heads is more likely.

Some Properties of the Entropy Function

The entropy function H(X) possesses several important mathematical properties that make it a robust and meaningful measure of information/uncertainty. These properties are not just theoretical curiosities they justify why entropy is the correct way to quantify information and guide the design of efficient coding and communication systems.

Property 1: Non- Negativity H(X) ≥ 0

Since $0 \leq P(x_i) \leq 1$ for all I , log□ $P(x_i) \leq 0$, making $-P(x_i)$log□ $P(x_i) \geq 0$. Therefore entropy can never be negative — uncertainty (or information) cannot be a negative quantity.

Example:

For any probability distribution, whether P = {0.5, 0.5} (H = 1 bit) or P = {0.9, 0.1} (H = 0.469 bits), the value is always ≥ 0.

Property 2: Entropy is Zero for a Certain Event

H(X) = 0 if and only if $P(x_i) = 1$ for some i (and 0 for all others)

If the outcome is completely certain, there is no uncertainty and hence no information gained.

Example:

A two-headed coin: P(Heads) = 1, P(Tails) = 0.

H(X) = −[1·log□1 + 0·log□0] = 0 bits (using the convention 0·log□0 = 0).

Property 3: Maximum Entropy for Equiprobable Outcomes

For a source with n symbols, H(X) ≤ log□ n, with equality if and only if all symbols are equally likely ($P(x_i) = 1/n$ for all i).

This is known as the Maximum Entropy Principle uncertainty is greatest when there is no bias toward any particular outcome.

Example:

For a fair 8-sided die, all outcomes have P = 1/8.

H(X) = log□8 = 3 bits (maximum possible for 8 symbols).

If the die were biased (e.g., one face more likely than others), H(X) would be strictly less than 3 bits.

Property 4: Symmetry

H(X) is a symmetric function of the probabilities P(x□), P(x□), ..., P(x□) it does not depend on the labels or order of the outcomes, only on the probability values themselves.

Example:

A source with P = {0.7, 0.2, 0.1} has the same entropy as one with P = {0.1, 0.7, 0.2} or {0.2, 0.1, 0.7}, since only the set of probabilities matters, not which symbol has which probability.

$H(X) = -[0.7\log 0.7 + 0.2\log 0.2 + 0.1\log 0.1] \approx 1.157$ bits in all three cases.

Property 5: Additivity for Independent Sources

If X and Y are two independent random variables, then H(X,Y) = H(X) + H(Y)

The joint entropy of independent sources equals the sum of their individual entropies combining independent uncertainties simply adds them.

Example:

Two independent fair coin tosses (X and Y):

H(X) = H(Y) = 1 bit each

H(X , Y) = H(X) + H(Y) = 2 bits

This matches direct calculation: 4 equally likely outcomes (HH, HT, TH, TT), each with P = 0.25, so $H(X , Y) = \log 4 = 2$ bits.

Property 6: Extremal Property (Grouping/Continuity)

Entropy is a continuous function of the probabilities a small change in $P(x_i)$ produces only a small change in H(X). It is also concave, meaning it increases as probabilities move toward uniformity and decreases as they move toward certainty.

Example:

Consider a binary source as p changes from 0 to 1:

p = 0.1 → H = 0.469 bits

p = 0.3 → H = 0.881 bits

p = 0.5 → H = 1.0 bit (maximum)

p = 0.7 → H = 0.881 bits

p = 0.9 → H = 0.469 bits

→ The entropy curve is smooth, symmetric about p = 0.5, and peaks at exactly 1 bit — illustrating both continuity and concavity.

Property 7: Entropy is Unaffected by Adding Zero-Probability Events

Adding an outcome with P = 0 to a distribution does not change entropy, since the term $-0 \cdot \log 0 = 0$ contributes nothing.

Example:

A die with an "impossible" 7th face (P = 0) has the same entropy as a standard 6-faced die: $H(X) = \log 6 \approx 2.585$ bits either way.

The communication system:

Information theory provides a mathematical framework to quantify, encode, and transmit information efficiently and reliably in communication systems.

Overview

Information theory, pioneered by Claude Shannon, is a **mathematical approach to understanding communication systems**. It focuses on the fundamental limits of information processing and transmission rather than the physical implementation of devices, making it central to both analog and digital communication systems. The theory quantifies information, measures uncertainty, and provides tools to design efficient coding schemes for reliable communication.

Key Concepts

Entropy (H): Entropy measures the **average information content or uncertainty** of a source. For a discrete source with symbols x_i occurring with probability p_i, Shannon defined entropy as:

$$H = -\sum_i p_i \log_b p_i$$

where b is the logarithm base, often 2 for bits. Higher entropy indicates greater uncertainty and more information per symbol.

Conditional Entropy: This measures the remaining uncertainty about a transmitted symbol after observing the received symbol, reflecting the effect of noise in the channel .

Mutual Information: Represents the amount of information shared between the source and the receiver, quantifying how much uncertainty is reduced by observing the output.

Source Coding: Also called data compression, source coding reduces redundancy in the transmitted data to achieve **efficient representation** without loss of information. The **source coding theorem** states that the average number of bits per symbol cannot be smaller than the source entropy.

Channel Coding: Introduces controlled redundancy to **improve reliability** over noisy channels. Channel coding allows detection and correction of errors, enabling communication close to the channel capacity.

Applications in Communication Systems

Information theory underpins the design of **digital communication systems**, including:

- **Determining channel capacity:** The maximum rate at which information can be transmitted reliably over a channel.
- **Error control coding:** Designing codes to correct errors caused by noise.
- **Bandwidth and power trade-offs:** Optimizing modulation schemes to balance signal-to-noise ratio and bandwidth usage.
- **Performance metrics:** Concepts like **ε-outage capacity** in fading channels help evaluate system reliability under varying conditions.

Practical Implications

By applying information theory, engineers can:

- Quantify the **minimum resources** (bits, bandwidth, power) needed for communication.
- Design **efficient coding schemes** to approach theoretical limits.
- Analyse **trade-offs** between reliability, speed, and resource usage in real-world systems.
  Information theory remains a cornerstone of modern communication, influencing everything from **telecommunications and data networks** to **wireless systems and storage technologies**. It provides a rigorous framework to understand and optimize the flow of information in complex systems.

In information theory, channel probability refers to the conditional probability $p(y \mid x)$that a communication channel outputs symbol $y$when symbol $x$is transmitted, forming the rows of a channel transition matrix that each sum to 1.

Channel probability:

Overview

Channel probability is a core concept used to mathematically model noisy communication systems. It defines how likely each possible output is for every possible input, capturing the uncertainty introduced by noise.

Key Components

- **Input alphabet** $\mathcal{X}$: Set of symbols that can be sent.
- **Output alphabet** $\mathcal{Y}$: Set of symbols that can be received.
- **Transition probabilities** $p(y \mid x)$: Conditional probability of receiving $y$given $x$was sent.
- **Channel matrix**: A stochastic matrix whose rows represent full probability distributions over outputs. Each row must sum to 1.

Discrete Memoryless Channel (DMC)

A DMC assumes:

- Symbols are transmitted one at a time.
- Each output depends only on the current input, not past transmissions.
- Forward channel probabilities $p(y_j \mid x_i)$define the channel completely.

Examples

1. Binary Symmetric Channel (BSC)

- Input/output: $\{0, 1\}$
- Crossover probability p: bit-flip probability.

2. Binary Erasure Channel (BEC)

- Output alphabet includes an erasure symbol “?”
- Example matrix includes probabilities of correct transmission, erasure, and incorrect output.

Backward Probabilities

Given a received symbol y, the probability that $x$was sent is:

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)}$$

This is essential for decoding.

Importance

Channel probabilities:

- Define how noise affects communication.
- Enable computation of **mutual information**, **conditional entropy**, and **channel capacity**.
- Form the basis for Shannon’s noisy-channel coding theorem, which determines the maximum reliable communication rate.

Joint and Conditional Entropies in Information Theory

Joint and conditional entropies are fundamental concepts in information theory that quantify the uncertainty in multiple random variables and the remaining uncertainty when some information is known. They are used to understand information flow, mutual information, and the chain rule that decomposes entropy into conditional components.

1. Joint Entropy

Joint entropy measures the total uncertainty associated with two random variables considered together.

For random variables $X$and Y:

$$H(X, Y) = -\sum_{x}\sum_{y} P\,(x, y)\log_2 P(x, y)$$

where:

- $P(x, y)$is the probability that $X = x$and $Y = y$occur together.
- $H(X, Y)$is measured in **bits** when logarithm base 2 is used.

Simple example

Suppose your project studies:

- X= Weather: Sunny or Rainy
- Y= Activity: Outdoor or Indoor

Joint entropy tells us **how uncertain we are about the combination of weather and activity**.

For example, possible outcomes are:

- Sunny + Outdoor

- Sunny + Indoor
- Rainy + Outdoor
- Rainy + Indoor

Higher joint entropy means the combinations are more unpredictable.

2. Conditional Entropy

Conditional entropy measures how much uncertainty remains about one variable after we know another variable.

The conditional entropy of Ygiven $X$is:

$$H(Y \mid X) = -\sum_{x}\sum_{y} P(x, y)\log_2 P(y \mid x)$$

It can also be written as:

$$\boxed{H(Y \mid X) = H(X, Y) - H(X)}$$

Example

Using the same example:

- X= Weather
- Y= Activity

Before knowing the weather, we may be uncertain whether a person will choose an indoor or outdoor activity.

After we learn that **it is raining**, we might predict "Indoor" more confidently.

Therefore:

Conditional entropy represents the uncertainty remaining in Activity after Weather is known**.**

3. Difference Between Them

Concept	Meaning	Formula
Entropy $H(X)$	Uncertainty in one variable	$-\sum P\,(x)\log_2 P(x)$
Joint Entropy $H(X, Y)$	Uncertainty in X and Y together	$-\sum\sum P(x, y)\log_2 P(x, y)$
Conditional Entropy $H(Y \mid X)$	Uncertainty remaining in Y after knowing X	$H(X, Y) - H(X)$

4. Important relationship

A useful equation for your project is:

$$\boxed{H(X, Y) = H(X) + H(Y \mid X)}$$

Think of it as:

Total uncertainty = uncertainty in X + remaining uncertainty in Y after X is known

Mutual Information

For your project, you can explain Mutual Information (MI) as a measure of how much knowing one variable tells us about another variable.

1. Definition

Suppose we have two random variables:

- X= Weather
- Y= Activity

Mutual Information measures how much knowing the weather reduces our uncertainty about the person's activity.

It is represented as:

$$\boxed{I(X;Y)}$$

The formula is:

$$I(X;Y) = \sum_{x} \sum_{y} P\,(x,y) \log_2 \left(\frac{P(x,y)}{P(x)P(y)} \right)$$

where:

- $P(x,y)$= joint probability of $X$and Y
- $P(x)$= probability of X
- $P(y)$= probability of Y

2. Simple Example

Consider:

- X = Weather: Sunny or Rainy
- Y = Activity: Outdoor or Indoor

Suppose people usually choose:

- Outdoor activities when it is sunny
- Indoor activities when it is rainy

In this situation, knowing the weather gives us useful information about the person's activity.

Therefore, Weather and Activity have mutual information.

If activity choices were completely unrelated to the weather, then:

$$\boxed{I(X;Y) = 0}$$

That means knowing Xgives us no information about Y.

3. Relation with Entropy

Mutual information can also be calculated using entropy:

$$\boxed{I(X;Y) = H(X) + H(Y) - H(X,Y)}$$

It can also be written using conditional entropy:

$$\boxed{I(X;Y) = H(Y) - H(Y \mid X)}$$

or:

$$\boxed{I(X;Y) = H(X) - H(X \mid Y)}$$

Easy way to remember

$$\boxed{\text{Mutual Information} = \text{Original Uncertainty} - \text{Remaining Uncertainty}}$$

For example:

$$I(X;Y) = H(Y) - H(Y \mid X)$$

Here:

- $H(Y)$= uncertainty about Activity before knowing Weather
- $H(Y \mid X)$= uncertainty about Activity after knowing Weather
- $I(X;Y)$= uncertainty removed by knowing Weather

4. Connection to Joint and Conditional Entropy

An important relationship is:

$$H(Y) = I(X;Y) + H(Y \mid X)$$

So you can think of it as:

Total uncertainty about Y = information provided by X + uncertainty still remaining

6. Application in Machine Learning

Mutual information is useful for **feature selection**.

For example, if we are predicting whether a student will pass an exam:

- X= Study hours
- Y= Pass/Fail

If knowing study hours provides useful information about the result, $X$and Ywill have mutual information.

So, in simple terms:

$$\boxed{\text{Higher MI} \Rightarrow \text{Stronger information relationship}}$$

$$\boxed{\text{MI} = 0 \Rightarrow \text{Independent variables}}$$

Encoding in Information Theory

For your project, you can explain **encoding** as the process of **representing information using symbols or binary bits (0s and 1s) so that the information can be stored or transmitted efficiently**.

1. What is Encoding?

In information theory, a source produces messages or symbols. **Encoding converts those symbols into codewords**, usually binary codewords.

For example:

Symbol	Binary Code
A	00
B	01
C	10
D	11

So if the message is:

A → B → C

the encoded message becomes:

00 01 10

2. Why Do We Need Encoding?

Encoding is mainly used to:

Represent information in digital form

- **Reduce storage requirements**
- **Transmit data efficiently**
- Help communication systems handle information systematically

3. Fixed-Length Encoding

In **fixed-length encoding**, every symbol gets the same number of bits.

Example:

Symbol	Probability	Code
A	0.5	00
B	0.25	01
C	0.15	10
D	0.10	11

Each symbol uses **2 bits**, regardless of how frequently it occurs.

4. Variable-Length Encoding

In **variable-length encoding**, different symbols can use different numbers of bits.

Frequently occurring symbols can be assigned **shorter codewords**, while less frequent symbols can receive **longer codewords**.

For example:

Symbol	Code
A	0
B	10
C	110
D	111

This idea can make encoding more efficient when some symbols occur much more frequently than others.

5. Connection with Entropy

This connects directly with the topics you've already covered.

Entropy tells us about the uncertainty or information contained in a source. Encoding tells us how we represent that information using bits.

So your project flow can be:

Entropy → Joint Entropy → Conditional Entropy → Mutual Information → Encoding

A simple way to remember it:

Entropy measures information, while encoding represents information.

6. Simple Real-Life Example

Imagine sending the word:

CAB

Suppose:

- A → 0
- B → 10
- C → 110

Then:

$$CAB \rightarrow 110\ 0\ 10$$

Therefore, the encoded bit sequence is:

$$\boxed{110010}$$

The receiver can decode the binary sequence to recover the original message.

CHAPTER-IV

FORECASTING

Forecasting is the process of predicting future events based on past and present data. It helps organizations estimate future demand, sales, production and business performance. Forecasting plays an important role in business planning, inventory control, finance, marketing, and decision-making.

Forecasting Methods:

Forecasting methods are the different techniques used to estimate future values based on historical data, statistical analysis, or expert opinions.

Forecasting methods are mainly classified into two categories:

- Qualitative methods
- Quantitative methods

Qualitative methods :

Qualitative methods are based on human judgement, experience and expert opinions. These methods are useful when historical data is not available such that as when introducing a new product into the market.

Quantitative methods:

Quantitative methods on the other hand. Use numerical data and mathematical models to predict future outcomes. These methods are more objective and provide accurate forecasts when sufficient historical data is available.

Quantitative forecasting includes techniques such as time series analysis , trend projection, moving average, exponential smoothing and casual forecasting .

Example:

A company planning to launch a new smartphone may ask experts and customers about future demand. This is a qualitative forecasting methods.

If the company predicts next year's sales using the predicts five year's sales data it is using a quantitative forecasting methods.

Components of time series:

A time series is a collection of observations recorded at regular

Intervals of time, such as daily, weekly, monthly or yearly. A time series consists of four important components:

- Trend
- Seasonal
- Cyclical
- Irregular variation

Trend Component:

The trend component represents the long-direction of data showing whether it is increasing or decreasing over time.

Seasonal Component:

The seasonal component refers to regular patterns that repeat during a specific season or period each year, such as increased sales during festivals.

Cyclic Component:

The cyclic component represents fluctuations caused by changes in business or economic conditions over several years.

Irregular component:

The irregular component includes unexpected variations caused by events such as natural disasters, pandemics, strikes or wars.

Understanding these components helps analysis identify patterns and make more accurate forecasts.

Smoothing method in forecasting :

Smoothing methods are forecasting techniques used to remove random fluctuations from time series data and reveal the underlying trend.

In practical situations, data often contains irregular ups and downs that make it difficult to identify the actual pattern. Smoothing methods reduce these random variations and make the data easier to understand. These methods improve forecasting accuracy by focusing on the general movement of data rather than temporary fluctuations.

Common smoothing techniques include the moving average method, weighted moving average method and exponential smoothing methods. These methods are widely used for short-term forecasting because they are simple to apply and provide reliable results

Moving average method:

The moving average method is one of the simplest and most widely used forecasting techniques. It predicts future value by calculating the average of the most recent observation in a time series.

As new data becomes available, the oldest observation is added.Therefore the average keeps changing or “moving”, which is why it is called the Moving average method. The moving average method is mainly used for short-term Forecasting. It helps to remove random fluctuations from data and shows the overall trend more clearly. Instead of using only one previous value, this method considers the average of several recent value, making the forecasting more reliable.

For example if a company wants to forecast next month’s sales

Using a s. month moving average, it first calculates the average sales of

The last three month. When the sales data for the next month becomes

Available, the oldest month’s data is removed, the new month’s data is

Added , and a new average is calculated this process continues for

Every forecasting period.

Formula:

Moving average= sum of the most recent n observations/n

Where,

n= number of observations.

forecast= average of the most recent observations.

Advantages:

- It is simple and easy to calculate.
- It reduces random fluctuations in data.
- It helps identify the overall trend.
- It provides reliable short-term forecasts.
- It is widely used in business and production planning.

Disadvantages:

- It gives equal weight to all observations.
- It is not suitable for long-term forecasting.
- It does not respond quickly to sudden changes in data.
- Some past information is ignored when new observations replace old ones.

Weighted moving Average method:

The weighted moving average method is a forecasting technique used to predict future values based on past data. Unlike the simple moving average, which Gives equal importance to all previous observations, the weighted moving average Assigns different weights of each observations . usually , the most recent data is given A higher weight because It is considered wore important for predicting future demand Or sales older data is given lower weights. This method helps produce more accurate forecasts when recent trend have a greater influence on future outcomes. Weighted Moving Average is widely used in business, economic, finance, production planning, inventory management and sales forecasting.

Advantages:

- Gives more importance to recent data.
- Produces more accurate forecasts when recent trends are significant.
- Reduces the effect of random fluctuation.
- Easy to understand and apply.
- Useful for short-term forecasting in business and industry.

Disadvantages:

- Choosing appropriate weights can be difficult.
- Incorrect weights way reduce forecasting accuracy.
- Not suitable for long-term forecasting.
- Does not handle sudden seasonal or irregular changes effectively.

Exponential smoothing method:

Exponential smoothing method is one of the most widely used forecasting techniques for predicting future values based on past observations. It is a simple and effective method that gives more importance to the most recent data while still considering older observations. Unlike the moving average method ,exponential smoothing does not require storing a large amount of historical data. Instead it updates the forecast every time new data becomes available.

Formula:

$$F_{t+1} = \propto A_t + (1-\propto)F_t$$

Where,

F_{t+1} = Forecast for the next period.

A_t = Actual value of the current period.

F_t =Forecast of the current period.

$\propto$ = smoothing constant (0 to 1)

Advantages :

- Simple and easy to calculate.
- Gives greater importance to recent observations.
- Requires only the latest forecast and actual value.
- Reduce the effect of random fluctuations.
- Helps Organizations makes better productions and inventory decisions.

Disadvantages:

- Not suitable for data with strong seasonal or long-term trends unless advanced smoothing method are used.
- May not respond quickly to sudden changes if $\propto$ is very small.
- Requires an initial forecast value.

Trend projection in forecasting:

Trendprojection in one of the simplest methods of time series in forecasting. It assumes that the past pattern or trend in the data will continue into the future.

1. Core idea

If sales, demand, population, etc have been consistently increasing or decreasing over time, we fit a line / curve to that historical data and extend it to predict future values.

The basic equation is :

y_t= a+bt+€t

Where,

y_t = forecasted value at time t

t = time period: 1,2,3,4,…

a= intercept the starting value when t=0

b= slope, the average change per time period

$€_t$= error term

Types of trend:

1. Linear trend:

Constant amount of increase/decrease each period.

Example: sales go up $5000 every year.

Formula : y_t= a+bt

2. Non-Linear trend:

a. Quadratic : y_t= a+bt+ct^2→growth rate is changing.
b. Exponential:y_t= a.b^t→constsnt % growth.

Ex: population, investment

c. Logarithmic: fast growth at start, then slows down.

3. No trend:

Data fluctuations around a flat average.

How to fit a and b

We usually use the least square method,

$$b=\frac{n\sum ty\sum y\sum}{n\sum t-(\sum t)}$$

a=y-bt

Example:

Year	T	Sales in 000
2021	1	10
2022	2	12
2023	3	14

Sales increases by 2 every year

So b= 2,a=8

Trend equation: y_t=8+2t

Forecast for 2025, t=5, y_5=8+2(5)=18→ $18,000

5.Advantages:

- Simple to calculate and explain.
- Good for long-term planning: capacity, budgets, strategy.
- Works even with limited data

6.limitations :

- Ignores seasonality and cycles→won't capture festival spikes, recessions.
- Assume trend continues →breaks if market condition change.
- Can be unrealistic long-term →exponential trends can't grow forever.

Seasonal components of forecasting:

Seasonal components= regular, repeating pattern that happens in fixed intervals.

Ex : ice cream sales go up every summer, sweater sales go up every winter.

It's one part of the 4 components of a time series.

The 4 components of a time series:

$Y_t=T_t+S_t+C_t+I_t$ Additive Model

$Y_t=T_t*S_t*C_t*I_t$ Multiplication Model

Component	Full form	Pattern	Example
T= Trend	Long-term direction	Up/down over years	Population growth
S=seasonal	Repeats every fixed period	<1 year cycle	Monthly, quarterly, Weekly
C=cyclic	Long waves >1 Year	Economic boom/ recession	5-7 years business Cycle
I= Irregular	Random Noise	Can't predict	Covid, strike, Earthquake

Seasonal is predictable and repeats. Irregular is random.

a. additive seasonality:

formula: $T_t+S_t+C_t+I_t$

Quarters	Avg sales	Seasonal index
Q1	80	80%
Q2	90	90%
Q3	110	110%
Q4	120	120%

Ex : store gets +500 customers every December regardless of sales level.

d. Multiplicative seasonality

$$Y_t=T_t*S_t*C_t*I_t$$

Seasonal fluctuations grow as trend grows

Ex : sales are 20% higher in December . if base sales = 1000→+200.

If base = 5000 → +1000, most business data uses multiplicative

3. seasonal index / seasonal factor

This how much higher / lower is this period us average".

Seasonal index of period = average for that season / grand away * 100

Index = 100→average season

Index = 120→20% above average

Index = 80 → 20% below average.

Ex : Quartely sales

Interpretation : Q11 is 20% above average year

4. how to forecast with seasonality:

2 main methods,

Method A :

Seasonal Decomposition + trend projection.

Step:

i) Find trend : remove seasonality using moving average.

ii) Fit a trend line $T_t=a+bt$

iii) Find seasonal indicates: $S_t=Y_t/T_t$ for multiplicative

iv) De seasonalize data: Y_t/T_t

v) Forecast trend for future periods.

vi) Re seasonalize : forecast = $T_{future} * S_{that\ season}$

Method : B seasonal models

i) Seasonal naïve : next Jan= last Jan . $F_t/m = Y_t$

ii) Seasonal exponential smoothing : Holt- winter's method. Handles trend + seasonality.

iii) SARIMA : seasonal ARIMA most powerful for complex seasonality.

5. numerical example:

Quarter	Year 1	Year 2
Q1	60	72
Q2	80	96
Q3	100	120
Q4	120	144

Step 1: average of Q1= 60+72/2=66

Grand average = 100

Step 2: Q1 seasonal index= 66/100-100= 66%

Similarly ,Q2=88% , Q3=110%, Q4=132%

Step 3 : Trend is increasing by 12 per year. Forecast 2026 trend = 140

Step 4 : forecast + Q1 2026 = 140 – 0.66 = 92.4

6. how to identify seasonality

- Plot the data →look for peaks every 12 months, 4 quarters , 7 days.
- Seasonal plot → plot all Jan together , all Feb together.
- Autocorrelation →plot spikes at lag 12 for monthly data.

7.advantages of modelling seasonality:

- More accurate forecasts for retail , tourism, agriculture.
- Better inventory planning→stock before season.
- Better staffing & budgeting .

8.limitations

i) assume pattern repeats→breaks if consumer behaviour changes.

ii) needs 2+ years of data minimum to get good seasonal indices.

iii) doesn't handle moving holidays like Easter , Diwali.

Casual forecasting method:

Month	Ad spend X in $1000	Sales Y in $'000
JAN	10	50
FEB	12	55
MAR	14	60
APR	16	65

Sales = a+b (Ad spend)

Forecast Y by finding what causes Y to changes.

If we know what drives sales, and we can forecast sales.

Formula : $Y= F(X_1,X_2,X_3,....$

Main types of casual methods:

a. regression analysis-most common

finds a mathematical relationship between what you want to forecast and what causes it.

1. simple linear regression cause:

$Y=a+bX$

Ex : ice cream sales= 200+15*(temperature) for every 1'c increase, sales go up by 15 units.

2. multi regression – many causes

$Y=a+b_1X_1+b_2X_2+b_3X_3$

b. Economic models:

A system of multiple regression equations used by government/banks.

Ex :GDP depends on consumption , Investment, Govt spending. And each of those depends on other things.

b. leading indicator app
c. use variables that change before Y changes.

Ex :building permits →predict housing construction 6 months later.

2.steps to build a casual forecast:

- Define the problem: where we forecasting
  Ex :monthly sales.
- Identity casual factors: what drives it.
  Ex: price, advertising , competitor price, season.
- Collect data: need historical data for both Y and all X's.
- Build the model: run regression to get equation check R^2.
- Forecast the X's: first forecast all the cause variables.
- Plug into equation get forecast for Y.

Ex for simple regression :

Data : Ad spend vs sales.

Month	Ad spend X i$000	Sales y in$000
1	10	50
2	15	65
3	20	80

Regression gives :Y=20+3X

Interpretation :

Base sales with 0 ad =20

Every $1000 extra ads →+$3000 sales.

Forecast : if next month Ad spend=$25,000

Sales : 20+3(25)=95→$95,000.

4. Advantages:

More accurate when you have strong casual factors.

Disadvantages:

- Hard to forecast X – your forecast is only as good as your X forecast.
- Assume relationship stays constant.

CHAPTER-V

EXAMPLES

Example sum:1

The function h(p) satisfying axioms 1 to 4 above can only be of the form

h(p)=-c log p $0 < p \leq 1$,

where $c > 0$ is an arbitrary constant.

Solution:

Let $y_1 = -\log_2 p_1$

$$y_2 = -\log p_2$$

Since

$p_i = e^{pi}$,

Put

g (y_i) = h (e^{yi})

there the additive property becomes

g $(y_1 - y_2)$ = g (y_1) = g (y_2)

also, form the monotonicity axiom,

g $(y_1) <$ g (y_2) , $0 \leq y_1 < y_2$

putting $y_1 = y_2 = 0$,

g(0) = 2 g(0)

therefore,

g(0) = 0

since g(0) < g(1) , c > 0

using g (y_1+y_2)= g(y_1)+ g(y_2)

we get

g(2) = g(1) + g(1) = 2c

g(3)= g(2)+ g(1) = 3c

similarly,

$$\therefore g(n) = nc,$$

Where n is an integer.

For a rational number

y= $\frac{m}{n}$

we have ng($\frac{m}{n}$) $= g(m) = mc$

hence, g($\frac{m}{n}$) $= \frac{m}{n}c$

thus , g(y)=cy

since y=-log p,

we get h(p)=g(-log p)

therefore,

h(p)=-c log p, c < 0

hence proved.

Example sum:2

Shows that the entropy of the following probability distribution is $2 - \frac{1}{2^{n-2}}$

Events	X_1	X_2	…	x_i	X_{n-1}	x_n	$X_{n=1}$
Probabilities(p_i)	$\frac{1}{2}$	$\frac{1}{4}$	…	$\frac{1}{2^i}$	….	$\frac{1}{2^{n-1}}$	$\frac{1}{2^n}$

Solution:

Given, $p_{i=\frac{1}{2^i}}$, i=1,2,3,…….,n-1

$$p_{n=\frac{1}{2^{n-1}}}$$

Entropy is,

H=-$\sum_{i=1}^{n} p_i \log_2 p_i$

Since

$p_i = \frac{1}{2^i}$,

$$\log_2 p_i = \log_2(\frac{1}{2^i}) = -i$$

Therefore,

H=-$\sum_{i=1}^{n} p_{i(-i)}$

H=$\sum_{i=1}^{n} \frac{i}{2^i}$

Thus,

H= $\frac{1}{2} + \frac{2}{2^2} + \cdots \ldots \ldots . . + \frac{n-2}{2^{n-1}} + \frac{n-1}{2^{n-1}}$

Multiplying by $\frac{1}{2}$,

$\frac{1}{2}$H=$\frac{1}{2}$+$\frac{2}{2^2} + \cdots \ldots \ldots \ldots + \frac{n-2}{2^{n-1}} + \frac{n-1}{2^n}$

Subtracting ,

H$-\frac{1}{2}H$=$\frac{1}{2} + \frac{2}{2^2}$+…….$\frac{1}{2^{n-1}} - \frac{n-1}{2^n}$

Hence,

$$\frac{1}{2}H = 1 - \frac{1}{2^{n-1}} - \frac{n-1}{2^n}$$

And therefore,

H=2-$\frac{1}{2^{n-2}}$

Hence proved.

Example sum:3

Week	1	2	3	4	5	6	7	8	9	10
Sales('000 rupees)	8	9	8	10	9	12	13	14	12	15

Compute the 3-week moving average forecast for the following data, showing weekly sales in('000)rupees of a company for three years:

Find the forecast errors and mean square error. What is the forecast for week 11?

Solution:

Computation of the 3- week moving average is as follows :

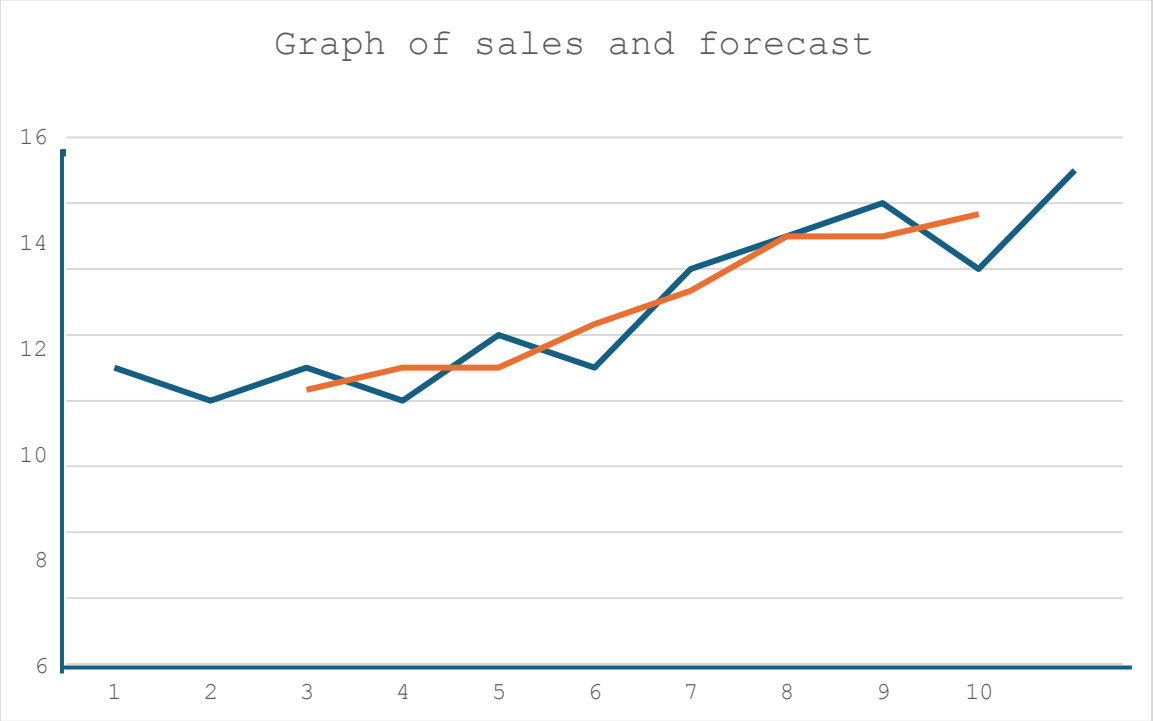
Week	Time series Value	3-week moving Total	Moving average forecast	Forecast Error	Squared forecast error
1	8				
2	9				
3	8				
4	10	25	8.33	1.67	2.78
5	9	27	9	0	0
6	12	27	9	3	9
7	13	31	10.33	2.67	7.13
8	14	34	11.33	2.67	7.13
9	12	39	13	-1.00	1
10	15	39	13	2.00	4
11	-	41	13.67	-	-

Mean square error= Total squared error/ number of forecast

= 31/7

=4.43

Forecast for week 11 is Rs.13670



Example sum : 4

Calculate 5-year moving average forecast for the following data using weights 1,1,3,2and1 respectively.

Year	1	2	3	4	5	6	7	8	9	10
Sales(Rs in lakh	40	33	72	81	76	68	91	87	98	97

What is the forecast for 11th year ?

Solution:

Calculation of weighted moving average forecast are under:

Sum of weight=1+1+3+2+1=8

Year	Sales (Rs.in lakh)	5-year weighted moving forecast
1	40	
2	33	
3	72	
4	81	
5	76	
6	68	(76×1+81×2+72×3+33×2+40×1)/8=65.875
7	91	(69×1+76×2+81×3+72×2+33×1)/8=71.000
8	87	(91×1+69×1+76×3+81×2+72×1)/8=76.000
9	98	(87×1+91×1+69×3+76×2+81×1)/8=78.750
10	97	(98×1+87×1+91×3+69×2+76×1)/8=85.125
11		(97×1+98×1+87×3+91×2+69×1)/8=89.125

From the above computations clearly the forecast for 11th year is Rs.8912500.

Example sum : 5

Calculate mean squared error (MSE)and mean absolute percentage error (MADE) for the forecast obtained in sample problem 3105 using $\propto$=0.8

Solution:

The above measures of forecast error are calculated in the following table :

Week (t)	Sales(in 1000)y_t	Forecast value$F_t(\propto = 0.8)$	$\mid y_t - F_t \mid$	$(\mid y_t - F_t \mid)$	$\frac{\mid y_t - F_t \mid}{y_t} \times 100$

1	113	108.0	5.0	25.00	4.42
2	101	112.0	11.0	121.00	10.89
3	98	103.2	5.2	27.04	5.30
4	107	199.0	8.0	64.00	7.47
5	120	105.4	14.6	213.16	12.16
6	132	117.1	14.9	202.0	11.28
7	110	129.0	19.0	326.0	17.27
8	117	138.8	3.2	10.24	2.73
9	112	116.4	4.4	19.36	3.93
10	125	112.9	12.1	146.41	9.68
Total	1209.22	85.13			

Using the above table, we get

MSE=1209.22/10=120.9

MAPE=85.13/10=8.5

Example sum :6

Demand (in ‘ 000 metric tonnes) for sugar of a sugar factory is given below:

Year	2004	2005	2006	2007	2008	2009	2010
demand	77	88	94	85	91	98	90

i) using the principle of least squares, fit a straight line trend. Show that the sum of deviations is equal to zero. Also determine the sum of squares of deviartions.

ii) Calculate the trend values and plot observed (actual)and trend values on a graph.

iii) Obtain the forecast of demand for the years 2012 and 2015.

Solution :

Let the straight line trend equation is given by

Where a and b are constants to be determined.

The given data is reproduced in the table below. Also , shown in the table is the calculation about the inputs for determining the values of a and b. notice that the years have been replaced by time scale with values 0,1,2,3,4 and so on.

Year	x (time)	Y (demand)	xy	X^2	y_t	$(y-y_t)$	$(y-y_t)^2$
2004	0	77	0	0	83	-6	36
2005	1	88	88	1	85	3	9
2006	2	94	188	4	87	7	49
2007	3	85	255	9	89	-4	16
2008	4	91	364	16	91	0	0
2009	5	98	490	25	93	5	25
2010	6	90	540	36	95	-5	25
Total	21	623	1,925	91		0	160

We have

$\sum y = na + b\sum x$=>623=7×a+b×21

$\sum xy = a\sum x + b\sum x^2$=>1925=a×21+b×91

And solving these two equations, we get

b=2 and a=83

i)The trend equation, therefore, is : y_t=83+2x. Using this equation, trend values for various year are obtained which are given in the 6_{th} column of the table. The next column gives the deviations of y-values from the corresponding y_t-values. Their sum is found to be equal to zero.

iii) Forecasting value:

y_{2012}= 83 + 2 x 8 = 99, (in '000 metric tonnes)

y_{2015}= 83 + 2 x 11 = 105,(in '000 metric tonnes).

Example sum: 7

A manufacture of optical lenses has the following data on the cost per unit (in Rs.) of a certain custom made lenses and the number of units made in each order:

Order no :	1	2	3	4	5	6
Number of units :	1	3	5	7	10	12
Cost per unit :	58	52	46	40	37	22

Obtained the regression equation on the cost per unit. Use this equation to estimate the cost of 15 units of lenses. Data intensive – need data for all X variables.

Solution.

Let the cost of an optical lense is denoted by y and the number of units of lenses by x. then we are to fit the equation of regression line by y

Order	X	Y	xy	X^2
1	1	58	58	1
2	3	52	156	9
3	5	46	230	25
4	7	40	280	49
5	10	37	370	100
6	12	22	264	144

.y_e= a+bx.

We have, $\bar{x} = \frac{\sum x}{n} = \frac{38}{6} = 6.33$

$\bar{y} = \frac{\sum y}{n} = \frac{255}{6} = 42.5$

$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\sum x^2 - n(x^2)}$

$= \frac{1358 \times 6.33 \times 42.5}{328 - 6 \times (6.33)^2}$

$= \frac{256.15}{87.58}$

$\bar{a} = \bar{y} - b\bar{x} = 42.5 - (-2.92)$

=42.5+18.40

=60.98

The regression equation y on x is given by

$$y_e = 60.98 - 2.92 \times x$$

$$y_e = 60.98 - (2.92) \times 5 = 17.8$$

Forecast x=15

Thus for 15 units of optical lenses, the cost of a lense is Rs.17.18